Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/01/2008
PI Name:	Long, Marshall Ph.D.		
Project Title:	Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment (NNX07AB33A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2006	**End Date:**	09/30/2009
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:			
Flight Assignment:	Flight definition NOTE: NEW grant, resulting from grants reinstatement directive in 8/2006, per D. Stocker/GRC (10/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell (Yale University)		
Grant/Contract No.:	NNX07AB33A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of NNC04AA03A, Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment, with grants reinstatement in 2006. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To better understand the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum - namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames which have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform from which to better understand the factors that affect diffusion flame lift-off distance and extinction and it should also provide an excellent test-bed to determine the roles of inception, surface growth and oxidation processes in soot formation models The goals of the research will be the generation of modified kinetic mechanisms for hydrocarbon fuels that are able to model effectively diffusion flame structure under a larger parameter range than existing mechanisms and the development of submodels for soot formation that are capable of preditcting both high and low soot loading levels in hydrocarbon flames of various fuels (methane, ethylene and propane).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place.
Task Progress:	In a follow-up to our previous NASA-funded work, we have continued the measurement of sooting flames at normal gravity. We have also worked to further validate our soot formation model by studying the interdependency of soot and NOx formation. More recently, we have shifted our focus to the development of flame configurations and diagnostics that will allow us to take advantage of the long periods of high-quality microgravity that are available on the International Space Station (ISS). These conditions will be critical to the study of both dilute and sooting flames. Microgravity Modeling The computational models used in our early flame studies included the chemical mechanisms of both GRI Mech 2.11 and a simple 26-species, C2 hydrocarbon mechanism developed at Yale. Both mechanisms clearly showed that buoyancy plays a role in both the size and shape of the laminar diffusion flame. Good comparisons between computations and experiments were obtained using the 26-species C2 mechanism for our standard, moderately diluted flame consisting of 65% methane and 35% nitrogen. However, the agreement in lift-off heights deteriorated rapidly as the dilution level in the flame was increased. Analysis of this disagreement pointed to a need for improved chemical kinetics. A recent computational effort re-analyzed our standard flames at both normal and zero gravity using a more up-to-date kinetics scheme based on a 66-species C2H4 mechanism. Analysis of other dilution levels and the incorporation of our soot model into the calculations are currently underway. Another area that requires further study in a microgravity environment is at the other end of the fuel dilution spectrum – namely pure fuel, sooty flame. Despite the complexities of modeling soot formation in flames using detailed chemistry, the linkage of soot production to radiation and bulk flame properties is so strong that the coupled treatment of this problem has become a necessity for quantitative modeling of flame structure. Recently, we have made significant progress related to the modeling of soot formation in diffusion flames, as outlined above. Computations agree well with experiments at normal gravity. Microgravity diffusion flames are of direct relevance to the study of soot formation/oxidation in turbulent diffusion flames as these systems are generally non-buoyant due to the large flow velocities. Additionally, microgravity flames have a significantly higher propensity to soot due to increased residence times. Consequently, a study of sooting tendencies at microgravity would provide invaluable information that can be used to improve further our soot formation model. We have begun implementing the soot growth model in our “zero-g” coflow diffusion flame program. Previous soot computations for methane were undertaken with an older version of our methane reaction mechanism and it is our goal to update the computations with GRI 3.0 and/or San Diego Mech. This requires that additional species and pressure dependent reaction rate coefficients be implemented within the program. In addition, the size of the fuel tube and the coflow will need to be changed to accommodate the constraints imposed by the ISS (see below). These modifications will require a multiple-step process be implemented to generate converged solutions for the new chemical and physical specifications. Microgravity Experiments A number of factors were considered in planning the experiments to be run on the ISS. The burner configuration and experimental plan must be compatible with the CIR. The burner used in previous studies required flow rates as high as 43 liters/min for the coflow, and 0.3 liters/min for the fuel flow. Due to the limited amount of gases available on the ISS, it is desirable that the experiment involve as low flow rates as possible. However, simply decreasing flow rates while using the same burner as in our previous experiments would result in smaller lift-off heights, increasing heat transfer from the flame to the burner (which is difficult to quantify and model). Instead, we have proposed reducing the diameter of our normal burner by a factor of two, making it possible to reduce flow rates significantly while maintaining a flame that is similar in composition to the flames imaged in earlier studies. In its proposed configuration, the burner will have a coflow with an outer diameter of 25 mm and a circular central tube of 2.1 mm inside diameter with a wall thickness of 0.13 mm. As a proof of concept we have constructed a smaller burner, taking note of the above requirements. In an effort to map out the range of flames to be studied at microgravity, we have begun to make measurements of normal gravity flames

	using this half-scale coflow burner. Both CH4 and C2H4 flames have been studied under a range of dilution levels and flow velocities to produce the desired variation in flame characteristics. It is expected that a number of flow conditions unstable in normal gravity will be stable under microgravity conditions. To conserve gas supplies, it is also important that the changeover time between different flow conditions be minimized. The volume of the central fuel tube inlet plenum has therefore been minimized. Calculations of the time required to flush the volume of the fuel/inert mixture between the mass flow controllers and burner outlet when switching between flow conditions have been made. We have begun the specification of the measurement techniques to be used in the proposed microgravity measurements aboard the ISS. Techniques include color imaging, UV imaging of chemiluminescence, soot pyrometry for temperature and soot volume fraction, thin filament pyrometry for flame temperatures in nonsooting regions of the flame, and laser extinction measurements for soot volume fraction. Color photography will be used to monitor the appearance and stability of the flames at the various dilution levels. These images can be compared with the results of the computations. Using a relatively simple measurement technique of imaging line-of-sight flame emission using a color digital camera, it is possible to make a number of observations on the flame. Such things as flame shape and size, lift-off height, and qualitative distributions of CH* and soot can be observed.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Connelly BC, Smooke MD, Long MB, Hall RJ, Colket MB. "Computational and Experimental Investigation of the Interaction of Soot and NOx in Coflow Diffusion Flames." 5th US Combustion Meeting, University of California at San Diego, San Diego, CA, March 25-28, 2007. Proceedings of the 5th US Combustion Meeting, University of California at San Diego, San Diego, CA, March 2007. , Mar-2007
Abstracts for Journals and Proceedings	Smooke MD. "Computational and experimental study of soot formation in coflow diffusion flames (Invited talk)." Invited talk at Eastern States Section of the Combustion Institute, Technical Meeting, Charlottesville, Virginia, October 21-24, 2007. Proceedings, Eastern States Section of the Combustion Institute, Technical Meeting, Charlottesville, Virginia, October 2007. , Oct-2007
Abstracts for Journals and Proceedings	Connelly BC, Bennett BAV, Smooke MD, Long MB. "A Paradigm Shift in the Interaction of Experiments and Computations in Combustion Research." Eastern States Section of the Combustion Institute, Technical Meeting, Charlottesville, Virginia, October 21-24, 2007. Proceedings, Eastern States Section of the Combustion Institute, Technical Meeting, Charlottesville, Virginia, October 2007. , Oct-2007